

Gold Macro Intelligence & Backtesting Application

A production ready Python application that analyses, backtests, scores, and forecasts Gold (XAUUSD) using macroeconomic data from FRED and market data from Yahoo Finance. It produces an explainable Gold Macro Score, generates Daily / Weekly / Monthly / Yearly signals with confidence levels, runs a walk forward backtest, detects market regimes, and serves results through an interactive Streamlit dashboard and a CLI report.

Research & decision support tool only. Not investment advice.

Features

- Downloads FRED macro series + Yahoo Finance market data (with local caching)
- Computes a 0–100 Gold Macro Score from six weighted fundamental drivers
- Generates Daily / Weekly / Monthly / Yearly signals with confidence scores
- Runs a walkforward backtest (2005–present) with no lookahead bias
- Detects market regimes (RiskOn, RiskOff, Stagflation, Recession, etc.) and measures historical gold performance by regime
- Produces a plain language report with full explanations
- Interactive 7page Streamlit dashboard
- All results cached locally; auto refreshes on new FRED/Yahoo data
- Graceful offline fallback to cached/synthetic data

Live Deployment

The app is deployed as a web application via GitHub + Streamlit:

Setting	Value
Main module	Gold_trading/app.py
Entry point	streamlit run Gold_trading/app.py
Framework	Streamlit Community Cloud / virtual Streamlit
Data	FRED API (live) + Yahoo Finance (live)

Project Structure

Gold_trading/

- └─ **config.py** FRED API key, series maps, weights, thresholds, paths
- └─ **utils.py** zscores, rolling stats, trend/acceleration/momentum helpers
- └─ **data_loader.py** FRED + Yahoo download, caching, unified daily panel
- └─ **feature_engineering.py** Economic calcs, feature construction, regime indicators
- └─ **macro_score.py** 6factor weighted Gold Macro Score + classification
- └─ **signals.py** Daily/Weekly/Monthly/Yearly signal engine + confidence
- └─ **regime.py** Market regime detection + historical gold performance
- └─ **backtest.py** Walkforward backtest + performance metrics
- └─ **dashboard.py** Streamlit multipage dashboard (render () entry point)
- └─ **app.py** Entry point: dashboard under Streamlit, CLI report otherwise
- └─ **requirements.txt**
- └─ **README.md**

Core Logic

Data Sources

FRED series (via fredapi, falls back to the official requests API):

Variable	FRED ID
US Dollar Index (broad)	DTWEXBGS
10Year Treasury Yield	DGS10
CPI	CPIAUCSL
Core CPI	CPILFESL
Core PCE	PCEPILFE
Fed Funds Rate	FEDFUNDS
Unemployment Rate	UNRATE
Nonfarm Payrolls	PAYEMS
Industrial Production	INDPRO

- Yahoo Finance symbols: GC=F (Gold), DXY.NYB (DXY), TLT, GLD.
- A publication lag shift is applied to FRED series so stale future values are never used. The Yahoo DXY column is renamed to DXY_YH to avoid a column collision with the FRED DXY (DTWEXBGS).

Gold Macro Score (0–100)

Factor	Weight	Bullish for Gold when
Real Yield	30%	Real yields fall
DXY	25%	Dollar weakens
Fed Expectations	15%	Fed eases / rates fall
Inflation	10%	Inflation rises (hedge)
Labor Market	10%	Unemployment rises (easing expectation)
ISM Momentum	10%	Manufacturing contracts (stagflation hedge)

$$component_score = clip(50 + 25 * z(directional_change), 0, 100)$$

$$MacroScore = \sum (weight_i \times component_score_i)$$

Classification:

Score	Label
90–100	Strong Bullish Gold
75–89	Bullish Gold
60–74	Moderately Bullish
40–59	Neutral
25–39	Bearish
0–24	Strong Bearish

Regime Detection

Classifies each day into RiskOn, RiskOff, Inflationary Expansion, Deflationary Slowdown, Stagflation, Recession, Recovery using growth Industrial Production Index (INDPRO_YoY), inflation (CPI_YoY), policy (FEDFUNDS), and risk sentiment (gold volatility). Also computes average forward gold return per regime.

How to Take Trades on a Given Timeframe

The application generates four independent signals (Daily, Weekly, Monthly, Yearly). Each maps to a different trading style, holding period, and risk tolerance. Use the signal that matches your horizon — and combine them for higher conviction.

Timeframe implies Trading Style

Signal	Typical Trader	Holding Period	Instrument Style
Daily	Day / swing trader	1–10 trading days	Short-term futures (GC=F), CFD, ETF (GLD)
Weekly	Swing / position trader	1–4 weeks	Swing trades, ETF, options
Monthly	Position trader	1–6 months	Medium-term positions
Yearly	Macro / investor	6–12+ months	Strategic allocation, long-term holdings

Rule of thumb: trade the timeframe whose holding period matches your available time to monitor and your risk budget. A daily signal is not a reason to hold a 6month position.

Signal implies Action

Signal	Action (if trading that timeframe)
Strong Buy	Long / add to long (high conviction)
Buy	Long (normal conviction)
Neutral	Flat — no new position; tighten trailing stops on existing
Sell	Short / reduce long (normal conviction)
Strong Sell	Short / add to short (high conviction)

Never trade against a Strong opposite signal. If the signal is Strong Sell, do not buy for a bounce — wait for the signal to flip.

Entry Rules

1. Signal confirmation: the signal must be Buy or Strong Buy (for longs) / Sell or Strong Sell (for shorts).
2. Confidence threshold: only act when confidence $\geq 65\%$ (configurable). Below that, treat as a *watchlist*, not a trade.
3. Macro Score filter: for a long, prefer Macro Score ≥ 60 ; for a short, prefer ≤ 40 . A neutral macro score (40–59) lowers conviction even if price signal is strong.
4. Regime check: avoid countertrend trades in extreme regimes (e.g., do not short gold in Stagflation/RiskOff without strong confirmation).

Exit Rules

- Take profit: target a risk reward $\geq 2:1$ (e.g., stop at $1 \times$ risk, target at $2 \times$ risk). Optionally scale out at $1:1$ and $2:1$.
- Stop loss: set below/above the ATR (14) based stop: stop distance = $k \times \text{ATR}(14)$, with $k \approx 2.0$ for daily, ≈ 3.0 for weekly.
- Trailing stop: after price moves $1 \times$ risk in your favour, move stop to breakeven; then trail by $2 \times \text{ATR}$.

- Time stop: if price hasn't moved favourably within the timeframe's expected holding period, exit.

Position Sizing (Risk Based)

Size each trade so that a full stoploss hit risks a fixed % of capital (e.g., 1%):

$$position_size = (capital \times risk_per_trade) / stop_distance$$

Example: capital \$10,000, risk 1% (\$100), stop distance \$30:

$$position_size = 100 / 30 = 3.33 \text{ oz } (\approx 0.33 \text{ contracts of GC} = F).$$

Confluence (MultiTimeframe Alignment)

Score conviction by how many timeframes agree:

Aligned Signals	Conviction	Suggested Sizing
4/4 agree	Maximum	Full size
3/4 agree	High	75% size
2/4 agree	Medium	50% size
≤1 agree	Low / conflicting	Skip

For example: Weekly Buy + Monthly Buy + Yearly Buy = strong long bias implies take the long.

Risk Management Rules

- Max 1–2% risk per trade, max 6% total open risk.
- Daily loss limit: stop trading for the day at –3% equity.
- Max drawdown: pause and reassess if equity drawdown exceeds the backtest's max drawdown metric.
- Position limit: never exceed 2× normal size on a single signal.
- Volatility filter: if gold_vol20 is above its 1year median, reduce size by half.

Worked Example (Weekly Swing Trade)

- Signal: Weekly = Buy (score +40), confidence 78%, Macro Score 68 (Moderately Bullish), Regime = Inflationary Expansion.
- Setup: Gold \$2,500, ATR (14) = \$25 implies stop distance = $3 \times 25 = \$75$.
- Size: capital \$10,000, risk 1% (\$100) implies $\$100 / \$75 = 1.33 \text{ oz}$ (≈ 0.13 contracts).
- Entry: enter long at market.
- Stop: \$2,425 (below $3 \times \text{ATR}$).
- Target: \$2,650 (2:1 R/R) — scale out half at \$2,575.
- Trail: after +\$75, move stop to breakeven; then trail $2 \times \text{ATR}$.

Trade Log & Review

Record every trade: signal, timeframe, score, confidence, macro score, regime, entry/stop/target, Rmultiple, and outcome. Review weekly if the live win rate diverges materially from the backtest, recalibrate thresholds or rerun the backtest.

Important: This framework is a decision support guide, not financial advice. Always verify against your own risk policy and obtain Risk & Compliance signoff before deploying capital.

Signal Score, Mathematics & Rationale

The signal engine converts raw macro/market data into a numeric score S in roughly $[-100, +100]$, which is then mapped to Strong Buy / Buy / Neutral / Sell / Strong Sell. Below is the exact math and the reasoning behind each piece.

Generic Formula

$$S(tf) = \sum_i w_i \cdot [m_i \cdot \tanh(z_i(t))]$$

Symbol	Meaning
$S(tf)$	Signal score for a given timeframe (tf = daily/weekly/monthly/yearly)
w_i	Weight of component i (the weights in each timeframe sum to 100)
m_i	Amplification multiplier for component i (usually 1 for

	zscores, 5 for raw price/trend ratios)
$z_i(t)$	Standardised input for component i at time t
tanh	Hyperbolic tangent, maps any real number into (1, +1)

The Two Key Transforms

1. z-score standardisation, $z = (x - \mu) / \sigma$, μ = rolling mean, σ = rolling standard deviation (in backtest, rolling to avoid lookahead; in live mode, full sample).

Why: It puts heterogeneous inputs on a common scale. A 2% gold return, a 0.3% yield change, and a 4% inflation reading are all converted to "how many standard deviations from normal." Without this, a large number of series would dominate.

2. tanh squashing, $\tanh(z) = (e^z - e^{-z}) / (e^z + e^{-z})$

Output is bounded to (1, 1).

Why:

- Bounded contribution: each component can contribute at most its weight, keeping S within a defined range.
- Saturating nonlinearity: extreme values give *diminishing returns*, so a single runaway factor (e.g., a huge one-day spike) can't dominate the score.
- Smooth & continuous: produces a well distributed score (including a meaningful Neutral band). This was chosen over sign() which created a bimodal distribution with almost no neutral readings.

Confidence

- Confidence = $\text{clip}(50 + |S|, 50, 100)$, weekly/monthly/yearly
- Confidence = $\text{clip}(50 + |S|, 50, 100) \times \text{Volatility Penalty}$, daily

Why $50 + |S|$: the further the score from neutral, the more confident the signal. Capped at 100 (never overconfident) and floored at 50 (never claim <50%).

Daily volatility penalty: Volatility Penalty = $\text{clip}(1 - \text{vol20} / \text{median}(\text{vol20}, 252), 0.3, 1.0)$. High volatility implies lower confidence, because noisy conditions make short-term signals less reliable.

Classification

Signal Score S	Signal
$S \geq +50$	Strong Buy
$+20 \leq S < +50$	Buy
$-20 < S < +20$	Neutral
$-50 < S \leq -20$	Sell
$S \leq -50$	Strong Sell

Per Component Formulas & Rationale

Daily

$$\begin{aligned} S_{\text{daily}} = & 40 \cdot \tanh(z(\text{gold_ret20})) \\ & + 25 \cdot [5 \cdot \tanh((\text{GOLD} - \text{SMA50})/\text{SMA50})] \\ & + 15 \cdot [5 \cdot \tanh((\text{SMA50} - \text{SMA200})/\text{SMA200})] \\ & + 20 \cdot \tanh(z(\Delta\text{macro_score}, 20d)) \end{aligned}$$

Term	Weight	Why (direction)
20day momentum	40%	Strong recent price momentum implies bullish
Price vs SMA50	25%	Price above 50day trend implies bullish
SMA50 vs SMA200	15%	Golden cross structure implies bullish
Macroscore change	20%	Improving macro conditions implies bullish

m=5 on the trend ratios because they are naturally small (e.g., 2% = 0.02); the multiplier brings them to a comparable magnitude as zscores.

Weekly

$$\begin{aligned} S_{\text{weekly}} = & 30 \cdot [5 \cdot \tanh((GOLD - SMA200)/SMA200)] \\ & + 20 \cdot \tanh(z(\text{macro_score})) \\ & + 25 \cdot [5 \cdot \tanh(-\text{yield_chg3_smoothed})] \\ & + 25 \cdot [5 \cdot \tanh(-\text{dxy_ret60_smoothed})] \end{aligned}$$

Term	Weight	Why (direction)
Price vs SMA200	30%	Longterm trend structure
Macroscore level	20%	Elevated macro score implies bullish
Yield change	25%	Falling yields implies bullish (lower opportunity cost)
DXY change	25%	Falling dollar implies bullish

Monthly

$$\begin{aligned} S_{\text{monthly}} = & 20 \cdot \tanh(\Delta CPI_{YoY}, 12m) \\ & + 25 \cdot \tanh(-\Delta FEDFUNDS, 6m) \\ & + 20 \cdot \tanh(-\Delta INDPRO_{YoY}, 6m) \\ & + 35 \cdot \tanh(-z(\text{real_yield})) \end{aligned}$$

Term	Weight	Why (direction)
Inflation change	20%	Rising inflation implies bullish (inflation hedge)
Fed change	25%	Falling Fed Funds implies bullish (easing) implies- sign
Growth change	20%	Slowing industry implies bullish (stagflation/recession hedge) implies- sign

Real yield level	35%	Low/falling real yields implies bullish implies– sign
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Yearly

$$\begin{aligned}
 S_{\text{yearly}} = & 30 \cdot \tanh(-\Delta \text{INDPRO}_{\text{YoY}}, 24m) \\
 & + 25 \cdot \tanh(\Delta \text{CPI}_{\text{YoY}}, 24m) \\
 & + 25 \cdot \tanh(-\Delta \text{FEDFUNDS}, 24m) \\
 & + 20 \cdot \tanh(-z(\text{real_yield}))
 \end{aligned}$$

Term	Weight	Why (direction)
Growth cycle	30%	Slowing long cycle growth implies bullish
Inflation cycle	25%	Rising long cycle inflation implies bullish
Policy cycle	25%	Long cycle easing implies bullish
Real yield level	20%	Low real yields implies bullish

Why the Signs Are Set This Way (Economic Logic)

- Gold is a nonyielding, dollar denominated, inflation sensitive asset. The sign of each term encodes the directional macro relationship:
- Rising real yields raise the opportunity cost of holding gold implies bearish implies invert (–).
- A stronger dollar makes gold more expensive in other currencies implies bearish implies invert (–).
- Fed tightening (higher rates) is bearish; easing is bullish implies invert (–).
- Rising inflation is bullish as a store of value hedge implies no inversion (+).
- Weakening labour / slowing growth raises the odds of Fed easing and stagflation risk implies bullish for gold implies invert (–).
- Falling/contracting manufacturing (ISM proxy) signals stagnation or recession implies bullish (safe haven / stagflation) implies invert (–).

Calibration Note

Each timeframe's weights sum to 100, and because tanh is bounded, the theoretical score is bounded. In the current implementation the trend ratio terms carry an extra $m=5$ multiplier, which amplifies their influence (effective weights of 125 / 75 on the price trend terms). If you want a strictly bounded $[-100, +100]$ score, calibrate m so that $\sum w_i \cdot m_i = 100$ per timeframe. The classification thresholds are applied to the raw S , so the direction and relative ordering are unaffected by this scaling.

Disclaimer

This application is a research & decision support tool. It is not investment advice. Any trading use requires review and signoff by the relevant Risk & Compliance function, and signals should be paper traded before capital is deployed.